Scenario Building by Design: Pathways Towards Anticipative Collaborative Governance by Design

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Abstract
This article explores how design scenarios can support anticipation, collaborative governance, and democratic decision-making across sectors. In focus is the use of scenario building that is communicated through conceptual and process diagrams, as well as contextual situations of people participating with designed artefacts to illustrate how scenario building can encourage community engagement to address challenges like water security and stewardship. By building on established design practices and literature on scenarios in design, the article proposes 'Anticipative Collaborative Governance' as a model for creating long-term, systemic solutions through design. It further contributes to cross-sector collaboration by introducing twelve guiding principles to strengthen connections between anticipatory, systemic, and collaborative design.

Keywords
Scenario building
Systemic design
Anticipative collaborative governance
Design futures
Designing for democracy

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Introduction

When climate concerns in the commons, such as droughts, floods, water security and river health, challenge us to explore possible futures in the public-problem solving arena, then we need to consider cross-sectoral collaboration. This kind of collaboration needs to reach for democratic practices within socio-ecological systems contexts while reaching for possible futures through inclusive and diverse civic imaginaires (DiSalvo, 2022, p. 21). More recently, transdisciplinary inquiry and perspectives have started to shape the future-orientation of governance. This futuring perspective related to governance concepts are informed by strategic foresight, which centres around strategic planning in collaboration around public-policy discourse (Ramos, 2014; Candy, 2018). Recently, DiSalvo (2022, p. 184) explored cross-sectoral governance through a design lens, with an additional focus on scenarios in a socio-technical paradigm (2022, p. 58). Instead, I turn to the use of scenarios, specifically *Scenario Building by Design* (Raymond, 2024a), as a way to approach collaborative governance in a mode of anticipation and design.

In this article, I present this anticipative scenario-centric design approach in relation to collaborative governance to arrive at conceptual meaning-making among cross-sectoral actors to foster what (Greenwood et al., 2021) refers to as 'agreement seeking' to inform 'taking collective action' ahead of time. In doing so, closer ties are made to the role, use, and purpose of design artefacts and processes as a democratic inquiry (DiSalvo, 2022). I present this democratic design approach as a novel concept, which I term *Anticipative Collaborative Governance by Design*.

Therefore, I ask how can scenario building practices and anticipatory design be integrated to inform and enhance collaborative governance as an approach for proactive, cross-sectoral public engagement? In addressing this question, I make use of two linked approaches I have developed, namely *Design Scenarios* (Raymond, 2024a), specifically Scenario Building by Design (Raymond, 2024a), to explore these anticipative collaborative practices among diverse cross-sectoral actors. I use Scenario Building Cards which were developed through a co-design process in a Global South urban-river context (Raymond, 2024a). The cards serve as cultural, conceptual, and meditative artefacts aimed at fostering meaning-sharing and meaning-making, through the use of word-centric serious games which I present here across two workshops. These workshops led to the definition of twelve guiding principles for *Anticipative Collaborative Governance by Design*. To present this, I make use of a Compositional Methodology by Design Research (Raymond et al., 2022), in an approach based on qualitative design research. Through this methodology, I draw on design methods and techniques to arrive at inter-related research contributions which include diagramming, annotating, layering, and compositing.

Perspectives on Collaborative Governance

Collaborative governance can be understood as a practice that combines elements of both Social Science, in terms of understanding societal relationships, norms, and collective behaviours, as well as elements from Political Science, in terms of understanding governance structures, power dynamics, and public policy processes (Ansell & Gash, 2008). I consider the Social Science perspective and approach collective behaviour in relation to meaning-making and futures-making through a design inquiry.

Riel Miller (2018), a scholar in Futures Studies and Foresight, emphasises the importance of governance approaches that incorporate foresight and anticipation. Stuart Candy (2018) whose work centres around Futures Studies, Foresight, and Serious Games, particularly in the context of public policy, develops tools, methods, and perspectives that support strategic decision-making, enhance public participation, foster innovation in the face of uncertainty and change. Both Miller's and Candy's work have implications for a futures-orientation to collaborative governance, whereas in the field of design, the scenario-centric work of Manzini (2001) considers Design-Oriented Scenarios (DOS) in relation to Social Innovation by Design. For example, Manzini (2007) positions DOS in relation to Social Innovation to propose new governance structures through collaborative communities. Parallel to Manzini's view on new governance structures and collaborative communities, Akama and colleagues view the collaborative engagement of communities as democratising design through Co-Design processes (Akama et al., 2019). Here Co-Design is enacted through narrative-centric approaches that foster inclusive and representative futuring processes (Akama et al., 2019).

From a Systems Design perspective, I follow the work of Rounsevell and colleagues, who have a Socio-Ecological Systems framing and use scenario-centric inquiry in relation to collective decision-making processes (Rounsevell et al., 2021). With regard to collaborative governance, I refer to systemic and summative perspectives offered by Greenwood, Singer, and Willis (2021), as well as Swilling (2020), who positions and frames collaborative governance in a South African context. These collaborative governance perspectives are scaffolded around autonomous collective action, self-organisation of cross-sectoral actors, collective sovereignty, and narrative-centric agreement seeking processes.

When shaping collective futures with people in place, as has been my intent, Design-Oriented Scenarios offer a way to include cross-sectoral actors in the conceptual phase of an anticipatory design process (Raymond, 2024a). In relation to collaborative governance, Design-Oriented Scenarios offer a narrative-centric way towards inclusive and collaborative processes, cross-sectoral perspectives among several actors, shared decision-making, and future-oriented approaches (Raymond, 2024a). These relational perspectives and views frame what I have conceptualised as 'Anticipative Collaborative Governance by Design'.

Anticipative Collaborative Governance by Design should be separated from two similarly defined terms: Anticipatory Governance, which is located within strategic foresight and governance views (Ramos, 2014), and Anticipatory Collaborative Governance,

which arises from within urban foresight and governance views (Ravetz & Miles, 2016). Instead, Anticipative Collaborative Governance by Design draws on a relational futures-oriented framing towards collaborative governance by mode of Anticipation and Design, by making use of Design Scenarios (Raymond, 2024a). I refer to scenarios as being conceptualised as narrative-centric approaches, that inform the understanding of possible, probable, preferred, or projected futures (Raymond et al., 2022).

From a Futures Studies perspective, Anticipative Collaborative Governance by Design does consider social foresight (Toffler, 1970, p. 239) in a public-policy orientation as a way to collectively shape shared futures. From a Design Studies perspective, it approaches the designing of inclusive collaborative processes through the development and use of designed artefacts, by taking an iterative, problem and solution framing, and user-centred approach (Raymond, 2024a). From a Social Studies perspective, it is a process-oriented and participatory approach to decision-making and problem-solving that aims to address complex social issues through advance collective action (Greenwood et al., 2021). At the centre of Anticipative Collaborative Governance by Design are collaborative processes that are positioned in a futures view, by taking collective action in the present.

Research and Design Methods

In this article I make use of qualitative design research mixed methods from which I draw a heuristic thematic analysis (Weber, 2014) to present 'Anticipative Collaborative Governance by Design', as well as twelve guiding principles to support this novel concept. I also use a compositional methodology by design research positioned in a relational ontology (Santos, 2015) that permeates the inter-relations between design and research: methods, techniques, and tools (Raymond et al., 2022). This contributes to what we have outlined as the development of alternative, situated, cultural and non-representational approaches to scenarios (Raymond et al., 2022). In addition, the relational ontology emphasises the use of visualisation techniques, such as diagramming, compositional and relational moves, to highlight different ways of knowing by design. My work here centres around a research by design inquiry in which I focus on Scenario Building practices. Scenario Building helps participants visualise and discuss potential scenarios by encouraging them to consider future-oriented actions and outcomes in a structured but imaginative manner.

I make use of Scenario Building cards that were developed prior to the workshops indicated in this paper, with people in place through Co-Design methods and techniques by mode of design hackathon events (Raymond, 2023). The cards were developed through a translation process that involved refining and distilling pragmatic and theoretical insights from the progressive work done on the Design Future Lexicon by Morrison and Dudani (2023), into clear, approachable language, and game mechanics designed to encourage reflection and dialogue. Through the mechanics of card play — pattern matching, word associations, and narrative-centric

prompts — participants were guided into a form of social learning that facilitated deeper engagement with each anticipatory-design theme embedded in the cards.

On the basis of earlier work in a longer qualitative study (Raymond, 2024b), I reflect on two 2-hour activity workshops with hybrid and online formats. The workshop series was designed as an iterative process towards conceptualising the guiding principles of Anticipative Collaborative Governance. Each workshop phase aimed to incrementally build on insights gained from the workshop participants. The hybrid activity workshop consisted of local cross-sectoral actors that included indigenous community leaders, start-ups, NGOs and academic scholars, in a South African context. The online activity workshop consisted of international cross-sectoral actors who work in urban-river contexts and included NGOs, academic scholars, and government officials. In these workshop settings, participants engaged in anticipatory reflection by using scenario building to project ethnographic insights into possible futures. This process transformed the ethnographic observations into a futures-facing framework.

Virtual collaborative interfaces and platforms were used (Miro and Microsoft Teams), which allowed me to gather both screen recordings and contextual and situated screenshots of people in play using the Scenario Building cards. The screenshots are layered with annotations and reflections of participants during the activity workshops which were captured through screen recordings of the workshop sessions and translated and visualised by means of audio transcription. Thereby, the visual material signals self-reflective performative collaborations that are in a mode of non-representational design research methods.

Potentials of Using Scenario Building

Collaborative governance has been a term used to define collective action among cross-sectoral stakeholders that seeks to take collective action in relation to Socio-Ecological Systems concerns (Greenwood et al., 2021). As 'Anticipative Governance' is emerging as a new term, so too have the collaborative governance practitioners adopted a futures-orientation to develop and define anticipative approaches in relation to collaborative governance. Exploring such futures opens up to design, which presents cross-sectoral collaborative processes through Co-Design, in a mode of democratising design (DiSalvo, 2022) and anticipative design (Raymond, 2024a). I draw on what I have devised as a Design-Oriented Scenario Building Framework (Raymond, 2024a). This framework allows one to position Scenario Building for Design so as to inform this mode of anticipation, and turn it to collaborative governance. Fig. 1 shows a systemic set of relations showing how Scenario Building by Design inquiry connects to spatio-temporal orientations (Raymond, 2023) which, in turn, inform and update ways of shaping futures.

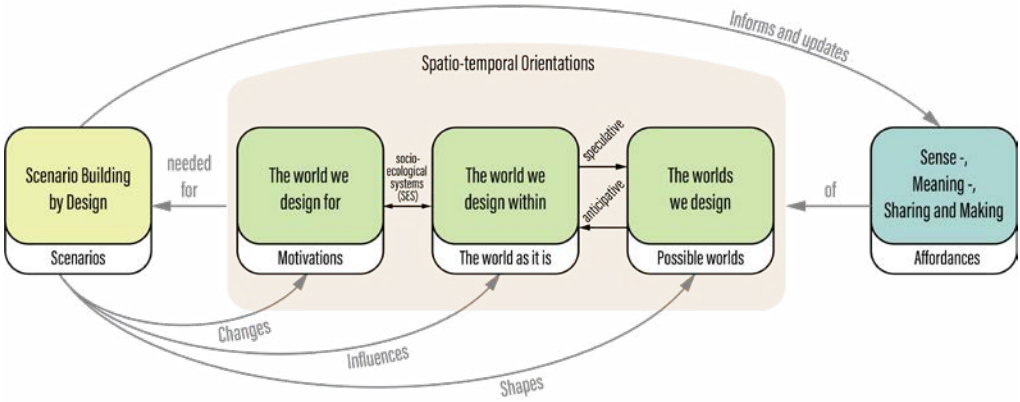


Fig. 1
A diagrammatic orientation to the role of scenarios in relation to shaping futures.

We can see, then, that Design Scenarios inform and update our affordances, such as sense, meaning, place, and futures making. These affordances are of possible worlds informed by mode of anticipation, as well as the world as it is informed by present-day socio-ecological systems concerns, and ultimately our motivations which are needed for scenario building by mode of scenario thinking (Raymond et al., 2022, p.3). In turn, Design Scenarios changes our motivations, influences the world as it is, and shapes possible worlds. In the following section, I will discuss how Scenario Building, through hybrid and online card-play in a mode of meaning-making through serious games, informs guiding principles towards my perspective on Anticipative Collaborative Governance by Design.

Scenario Building by Design Through Serious Games

Serious games and their context of use

Clark Abt (1987) first introduced the term Serious Games and says that when "reduced to its formal essence, a game is an activity among two or more independent decision-makers seeking to achieve their objectives in some limiting context" (p.9). Where serious games "[...] are not intended to be played primarily for amusement" (p.9). Serious Games are oriented towards education, training, health and public policy. It is in this public domain that we see Serious Games open up modalities of dialogue exchange, narrative-centrism and knowing, through play between several actors (Flanagan & Nissenbaum, 2014, p. 96). These modes of play are most notably characterised by card and canvas-based play (Raymond, 2024a; Candy, 2018).

Serious Games: Hybrid Workshop

I present an assemblage of screenshots captured during the hybrid workshop, carried out to clarify ethics in relation to the context of the inquiry. Fig. 2, depicts ten participants, composed of local community actors, start-ups, NGO's, researchers, and Indigenous group representatives in a South African context, who used Scenario Building cards to develop ethical guidelines for collaborative governance

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to inform a social impact framework. Phase 1 shows the workshop setting at Stellenbosch University's Water Institute (SUWI), where actors engage in a mode of meaning-sharing. The playing-face of each card consists of a key design-futures word and the definition of it that was borrowed from the Design Futures Lexicon (Morrison & Dudani, 2023), as well as a pattern-matching game mechanic that allows actors to play and match a card with one that has been played before (Raymond, 2024a). This has been designed to afford relational thinking, meaning-sharing and meaning-making, based on contiguous word associations (Raymond, 2024a). The cards also invite ways of knowing through modes of play, to discursively allow actors to voice their thoughts in challenging settings, as well as to invite the metaphoric act of playing the hand you are dealt.

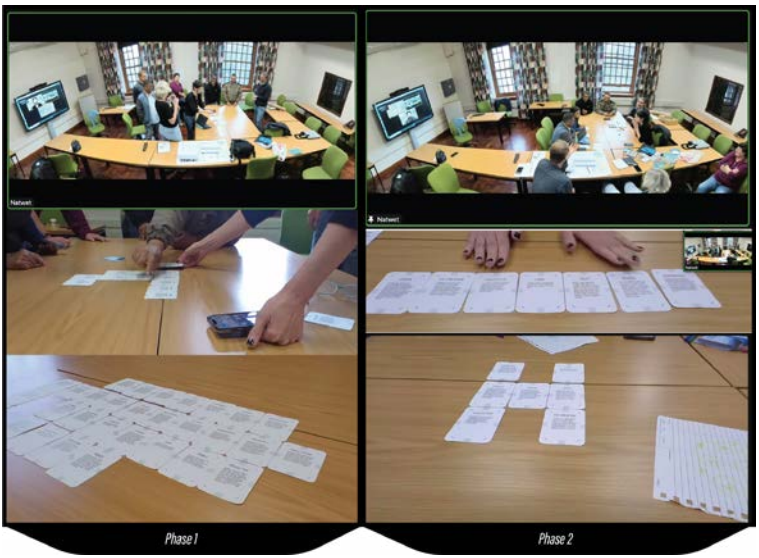


Fig. 2
Interactions during a two-phased activity workshop where actors are using the Scenario Building Cards.

Phase 2 indicates a more critical, deliberative, and negotiative exchange among participants where they are engaged in card sorting. The actors adapted and modified the playability of the cards in order to select and arrange seven concepts related to the ethics of collaborative governance and social impact projects.

Serious Games: Online Workshop

Following the hybrid workshop, an online activity workshop was conducted with four participants. Fig. 3 depicts two Miro-board canvases (phase 1 and phase 2) which were used to facilitate online play, as actors can virtually select, drag, and place objects. The canvas set was iteratively developed following reflections from the hybrid workshop, where actors could use the first phase to familiarise themselves with the gaming mechanics of the Scenario Building cards, and then transition to the second phase where they could rearrange the cards without prescribing to the pattern-matching gaming mechanic. This improved the general playability. Throughout the

online workshop, participants could make use of the native collaborative tools in Miro as well as see and talk with each other virtually over Microsoft Teams.

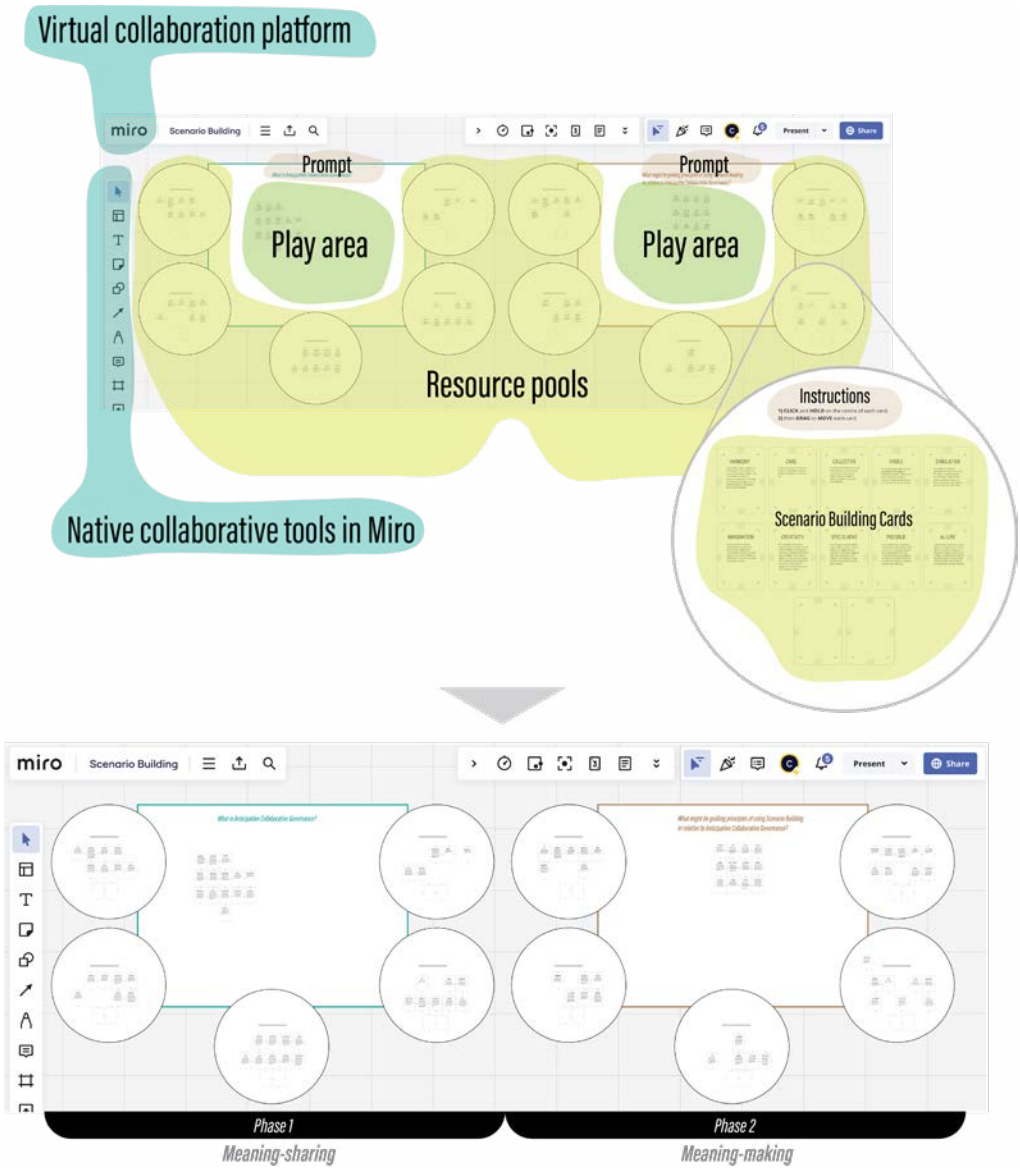


Fig. 3
Interactions during a two-Orientations to, and the design of, the canvases used during the online activity workshop.

Each canvas has an open-ended question (prompt), which actors had to deliberate and negotiate by playing the Scenario Building cards (from the resource pools into the play area by following the instructions). With regards to collaborative governance, this relates to collective agreement seeking (Greenwood et. al., 2021; Swilling, 2020). The first canvas presented participants with the question: “What is Anticipative Collaborative Governance?” This question was set to

open dialogue around the subject of the workshop, and to gather collective thoughts and opinions on how the actors may conceptualise Anticipative Collaborative Governance by Design using the Scenario Building cards.

The second canvas probed the question: "What might be the guiding principles of using Scenario Building in relation to Anticipative Collaborative Governance?" This question was followed with the prompt to use the Scenario Building cards in a mode of critical reflection. In addition, participants could also make use of Miro's sticky notes and type tools to transcribe any words, meanings and definitions that might not be featured on the Scenario Building cards. Next, I go into further detail about the contributions of the participants to the canvas space. First, however, I mention the importance of temporal-spatial aspects of the process and inputs.

Scenario Building cards in Use: Meaning-sharing

Tony Fry (2017, p.4) writes about ways of knowing in relation to the future and considers textual material as semiotics of meaning when considered in relation to design. By means of Serious Games, these exchanges of meaning between actors when playing the Scenario Building cards occur as chronotopes. Time (chrono) and topes (place) offer spatio-temporal exchanges in what Greenwood, Singer and Willis (2021) refer to as collective agreement seeking. It is by acknowledging that we think in terms of where we are at, that we may consider what Fry (2017, p. 12) describes as being "ontologically grounded in a particular ecology of mind, epistemological milieu, and changing sociopolitical and cultural environment". This meaning-sharing capacity among actors over space and time, using Miro as an online platform for hermeneutic exchanges, is what the first canvas is inviting.

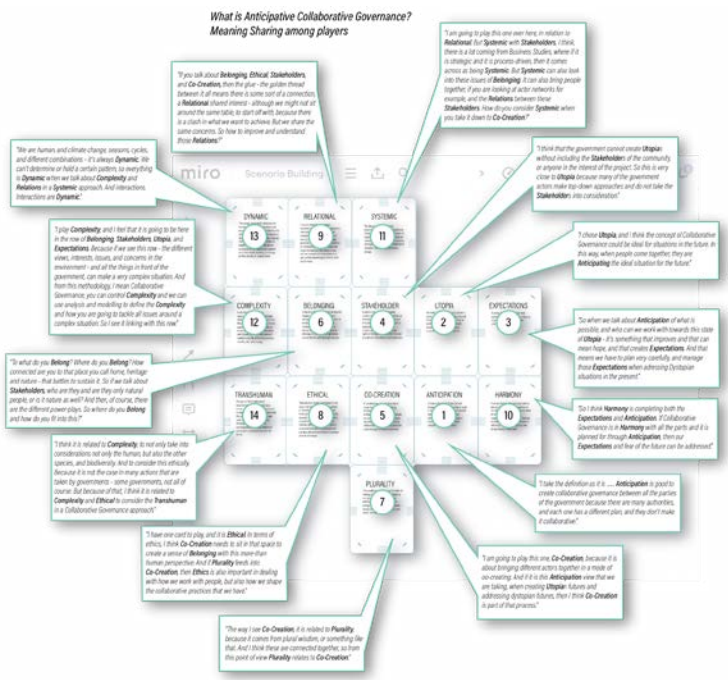


Fig. 4
An assemblage and sequencing of the Scenario Building cards played during phase 1 of the online workshop, layered with the transcriptions of actors while playing the cards.

Fig. 4 is an assemblage of the cards being played in the numerical order of their play. It is also layered with speech bubbles, used to capture transcripted expressions by each participant as they played the card in relation to one previously before . These relational exchanges of meaning are made visible by using bold text in the speech bubbles. The dialogical meaning-sharing taking place is exemplified by each player speaking from their own contextual orientation based on their interpretation of the word on each card. This contiguous relational framing is afforded by the pattern-matching semiotic on the border of each card. The gaming mechanics of pattern matching affords actors to express their diverse views through their own interpretation (Raymond, 2024a) and opens a space for the actors to assimilate with each other's world-views to define Anticipative Collaborative Governance.

Scenario Building Cards in Use: Meaning-Making

In *The Design of Everyday Things* Donald Norman (2013) emphasises the role of affordances in facilitating intuitive interactions between actors and designed objects. It is possible to see how participants during the second phase of the online workshop constructed, interpreted, and attributed meaning to the word-based Scenario Building cards. Here the cards afforded the personal experiences of the actors to actively construct meaning based on their own cultural, social, and personal contexts. These exchanges were dialogical and discursive and resulted in the participants arriving at an analytical framing of the guiding principles of Anticipative Collaborative Governance by Design.

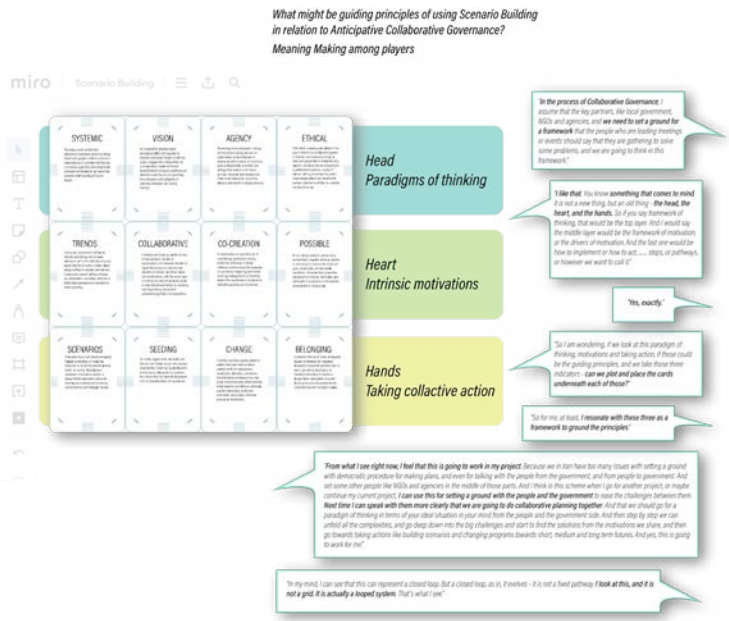


Fig. 5
An assemblage and grouping of the Scenario Building cards played during phase 2 of the online workshop, layered with the transcriptions of dialogue exchanges of actors.

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In Fig. 5, the Scenario Building cards have been assembled based on the prompt: What might be the guiding principles of using Scenario Building in relation to Anticipative Collaborative Governance? The text on the right is a transcription of the dialogue exchanges between participants, where they first deliberated their interpretation of the prompt, and by negotiating conflicting directions, ended up identifying a conceptual apparatus to ground the principles. First the actors identified 'Head', as a paradigm of thinking. Secondly, they identified 'Heart', which draws on intrinsic motivations. Lastly, they identified 'Hands', which they scaffolded around the collaborative governance concept of taking collective action (Greenwood et al., 2021). Within each of these three conceptual framings, the actors arranged the cards they considered to be most appropriate to act as guiding principles.

The two text blocks at the bottom are reflections after completing the activity of arranging the cards in a meaning-making mode, in which they share views of how to incorporate the guiding principles in their personal projects and domain fields. For example, in Fig. 5, one participant observes "I can use this for setting a ground with the people and the government". One participant offered a zoomed-out reflection on the conceptual apparatus by stating that "I look at this, and it is not a grid. It is actually a looped system". This reflects the views of Norman (2013) whereby actors intuitively engage in meaning-making by using designed objects. In this instance, these participants expressed cross-sectoral collaboration and systemic views related to Anticipative Collaborative Governance by Design.

Guiding Principles for Anticipative Collaborative Governance by Design

These guiding principles are informed by the Scenario Building cards that were intuitively played between the various actors. This interactive format of Serious Games shifted the social learning context by creating a space in which participants could iteratively build on one another's insights, sparking new understandings of collaborative governance concepts. Moreover, individual reflection was enhanced as participants were prompted to consider each card's meaning within their unique contexts. The process encouraged participants to connect abstract governance principles to their personal experiences, which resulted in more profound personal insights and a stronger grasp of the relevance of Anticipative Collaborative Governance by Design to real-world challenges.

The result of the activity workshop has been the construction and development of twelve guiding principles for Anticipative Collaborative Governance by Design. In terms of meaning-making, these twelve principles were derived from intrinsic broaching skills (Fry, 2017, p. 29), which were facilitated through the use of serious games as a mode of peer-play (Candy, 2018). Referring to Fry (2017), broaching skills act as communication techniques that foster open and respectful dialogue, allowing diverse perspectives to be expressed, understood, and integrated, especially in settings where participants may hold different or conflicting viewpoints. These broaching skills consist of effective communication, sensitivity,

context, active listening, and conflict resolution between the actors. These skills were necessary for navigating different viewpoints and aligning the principles with the core goals of Anticipative Collaborative Governance by Design.

Collaborative Governance is typically positioned within social and political science, and these twelve principles that were developed through design futuring enactments, through scenario building, explore anticipative meaning-making and meaning-sharing through Serious Games. Even though terms such as collaboration, design, systems, anticipation, scenarios, and governance are not new, these 12 principles, framed by these terms, offer a new perspective through which to reconstruct anticipatory ways of collaborating within real-world settings. For instance, 'Head', the paradigm of thought, comprises Systemic, Vision, Agency, and Ethical as principles. These principles were decided upon to inspire holistic thinking and collective goals that leave no being behind. 'Heart', the intrinsic motivations, comprises Trends, Collaborative, Co-creation, and Possible as principles. These principles were motivated by the participants to inspire trust through informed processes of futures-shaping. 'Hands', taking collective action, comprises Scenarios, Seeding, Change, and Belonging as principles. The participants viewed these principles as foundational to enacting futures. Altogether, the participants' intentions were not to generate a rigid conceptual apparatus, instead, one that is multi-scalar and holistic.

The twelve principles are collectively displayed in Fig. 6, which, through directional lines, indicates they may be further understood as a set of relationally linked guidelines. These principles may be read from left to right, top to bottom, right to left, and bottom to top in a poli-directional relational dynamic. Following is a list of the twelve principles based on the meaning-making exchanges between participants.



Fig. 6
Twelve guiding principles to Anticipative Collaborative Governance by Design as defined by the actors, as well as an abstraction of a meta relational orientation.

By grounding the principles in these themes (head, heart, hands), the workshop participants were able to generate a conceptual apparatus that is both comprehensive and adaptable by reflecting the shared insights across the two workshop phases.

Implications

The twelve principles presented in this paper hold significance as an orientation towards addressing socio-ecological systems concerns in the context of cross-sectoral collaboration within a governance framework. A possible use of the Anticipative Collaborative Governance by Design conceptual apparatus of Head, Heart, and Hands in cross-sectoral collaborations may be invoked by project teams that epistemically consist of scientific inquiry which aligns with head, public services which aligns with heart, and practitioners that aligns with hands. The challenge with this suggested cross-sectoral collaboration would be to create dynamic and interchangeable epistemic flow between all actors and not to necessarily conceive the Anticipative Collaborative Governance by Design conceptual apparatus as a tiered construct which may further embed power tensions within team dynamics.

In terms of Co-design, the Scenario Building cards were co-designed prior to the workshops presented in this paper; through the serious games, the participants collaboratively design the configuration and mapping of the cards during the meaning-sharing and meaning-making phases; and in this paper, the participants also co-designed the Anticipative Collaborative Governance by Design conceptual apparatus. As a novel concept that considers collaborative governance in relation to futuring approaches by design, Anticipative Collaborative Governance by Design offers a mode of inquiry by design that differentiates between the public-policy and future orientation presented by strategic foresight and urban foresight. The twelve principles offer a conceptual framing for taking collective action and could hold value for practitioners, scholars, and governing authorities to explore collaborative practices addressing public-problem solving concerns. The next step is to explore how these principles are applied in policy contexts as a bottom-up approach informed by design, governance, and anticipation. In addition, Anticipative Collaborative Governance by Design, as framed by Scenario Building cards, by means of serious games, affords ways to design for sustainability, as well as design for democracy, and advances the public-facing orientation of design futures.

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